

```
cat movie.mpeg.0* > movie.mpeg
```

```
cat > lazy_dog.txt
```

The quick brown fox jumps over the lazy dog.



Loading Memory

1- regular file

2- Directory

3- Socket

4- Symbolic link

5- Pip

6- dev

→ Terminal Emulator :-

- exit
- uptime
- df, du
- mount

mount → /dev/sda → /home/user

- free

du -sh → Print size of current Dir

When using a graphical user interface (GUI), we need another program called a terminal emulator to interact with the shell.

- sudo shutdown -n

- ls

- file

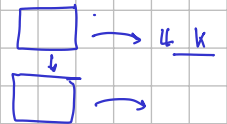
→ history → default → 1000 Command

→ ↑
↓

Key	Action
Ctrl-d	Delete the character at the cursor location.
Ctrl-t	Transpose (exchange) the character at the cursor location with the one preceding it.
Alt-t	Transpose the word at the cursor location with the one preceding it.
Alt-l	Convert the characters from the cursor location to the end of the word to lowercase.
Alt-u	Convert the characters from the cursor location to the end of the word to uppercase.

OS → File

Block



- 256 byte → 4k

Ctrl-a	Move cursor to the beginning of the line.
Ctrl-e	Move cursor to the end of the line.
Ctrl-f	Move cursor forward one character; same as the right arrow key.
Ctrl-b	Move cursor backward one character; same as the left arrow key.
Alt-f	Move cursor forward one word.
Alt-b	Move cursor backward one word.
Ctrl-l	Clear the screen and move the cursor to the top-left corner. The clear command does the same thing.

Key	Action
Ctrl-k	Kill text from the cursor location to the end of line.
Ctrl-u	Kill text from the cursor location to the beginning of the line.
Alt-d	Kill text from the cursor location to the end of the current word.
Alt-Backspace	Kill text from the cursor location to the beginning of the current word. If the cursor is at the beginning of a word, kill the previous word.
Ctrl-y	Yank text from the kill-ring and insert it at the cursor location.

Table 8-4: Completion Commands

Key	Action
Alt-?	Display a list of possible completions. <i>On most systems you can also do this by pressing Tab a second time, which is much easier.</i>
Alt-*	Insert all possible completions. This is useful when you want to use more than one possible match.

History

Table 8-6: History Expansion Commands

Sequence	Action
!!	Repeat the last command. It is probably easier to press up arrow and ENTER.
!number	Repeat history list item <i>number</i> .
!string	Repeat last history list item starting with <i>string</i> .
!?string	Repeat last history list item containing <i>string</i> .

each file has
 rwx rwx rwx
 7
 000
 111

> foo.txt //create empty

-	A regular file.
d	A directory.
l	A symbolic link. Notice that with symbolic links, the remaining file attributes are always "rwxrwxrwx" and are dummy values. The real file attributes are those of the file the symbolic link points to.
c	A <i>character special file</i> . This file type refers to a device that handles data as a stream of bytes, such as a terminal or /dev/null.
b	A <i>block special file</i> . This file type refers to a device that handles data in blocks, such as a hard disk, SSD, or DVD drive.

id
 -chmod
 -umask
 -su
 -sudo
 -chown
 -chgrp
 -addgroup
 -usermod
 -passwd
 -adduser
 -usermod -af
 -userdel -r

User	Group	Other
rwx	rwx	rwx

-rwx --- --
 -rw- r-- r--

chmod
 octal
 +r
 -w
 +rx
 +rwx

→ User Management

- each user has line in /etc/passwd
- has name, passwd, uid, gid
- homedirectory
- login shell

create user using

`sudo useradd Zanksha`

- UID will assigned to Zanksha
- /etc/passwd Append new line
- Create empty home directory
- Shell /bin/bash

`sudo useradd` ^{UID} `-u 1005` ^{GID} `-g 1002` ^{home dir} `-d /home/Zanksha` ^{default shell} `-s /bin/sh` ^{Password} `-p "123"` ^{comment} `-c "Zanksha Your Pass 123"`
`Zanksha`

→ Delete user:

`sudo userdel <username>`

all references related user will be removed except:

- ① /etc/passwd ✓
- ② /etc/group ✓
- ③ /etc/gshadow ✓

to solve this problem

`sudo userdel -r` ^{recursive}

Modify a user account

`$ sudo usermod <options> <username>`

open question

- a -- add the user to the supplementary group(s)
- G -- a list of supplementary groups which the user is also a member of
- l -- new value of the login name
- L -- lock the user account
- U -- unlock the user account
- d -- new home directory for the user account
- m -- move the content of the user's home directory to the new location
- s -- new login shell for the user account
- p -- new encrypted password of the user account
- c -- new value of the user's password expiry date

Open Question:-

① → supplementary

② → what happen when create user without password

③ /etc/sudoers → rules that users must follow when using sudo command.

④ slow symbolic link & fast symbolic link

Creating groups using groupadd

\$ groupadd testgroup -- creates a group called testgroup

Deleting groups using groupdel

\$ groupdel testgroup

Adding users to groups using usermod

\$ usermod -aG <groupname> <username>

Id

sd

Any Linux File = Filename + inode + datablocks

Linux OS use Inode number Not name

→ Inode is an internal storage associated with each file contains the following info:

- 1- Permission
- 2- number of Link
- 3- User, group owner
- 4- File size
- 5- time stamp
- 6- Pointer to data block

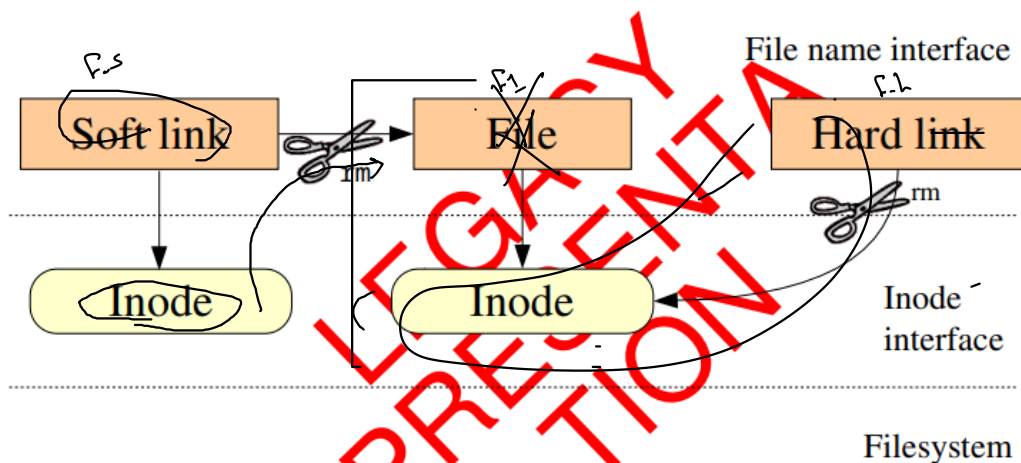
} struct

block 

Link

Reference (Pointer) to File

ls -li → inode number
stat → File status



`ln -s src ds` → hard Link

`ln -s` → soft link

```
eslam@eslam:~$ ln f1 f_h
eslam@eslam:~$ ln f1 f_s -s
eslam@eslam:~$ ls -li f1 f_h f_s
8948106 -rw-rw-r-- 2 eslam eslam 0 Jan 23 20:15 f1
8948106 -rw-rw-r-- 2 eslam eslam 0 Jan 23 20:15 f_h
8947388 lrwxrwxrwx 1 eslam eslam 2 Jan 24 22:25 f_s -> f1
```

- hard link

- Not new file
- Same file content
- Same Inode
- additional data entry with different filename / path same inode

`rm h1` → remove the original file

`unlink <sym/hardlink>`

the default behavior for `ln` is to create hard links

A hard link to a file is a regular file with exactly the same physical contents

While they still save space, hard links can't be distinguished from the original files.

If you remove the original file, there is no impact on the hard link contents.

The contents are removed when there are no more files (hard links) to them